Process FMEA (PFMEA)

Find where each process step could fail, how bad that would be, and what stops it, before it happens on the line.

FMEA NUMBER	PROCESS OR LINE	PART NUMBER AND REVISION

PREPARED BY	TEAM	DATE / REVISION

How to use this template

1. Take the steps from your process flow, in order. For each step, ask how it could fail to meet its requirement.
2. For each failure mode, write the effect on the customer, the cause, and the controls you have today. Then score S, O and D using the scales below.
3. Prioritise by RPN (S × O × D), or by Action Priority if your customer follows the AIAG & VDA FMEA Handbook. Always act on high severity first, whatever the total.
4. Re-score after each action, and keep the PFMEA in step with your control plan and work instructions. There are 75 rows over five worksheet pages; for a longer process, save a second copy.

S Severity, 1 to 10

How bad is the effect if it happens? 10 = safety or legal risk without warning. 7–8 = product doesn't work, customer very unhappy. 4–6 = works but degraded. 1 = no noticeable effect.

O Occurrence, 1 to 10

How likely is the cause, with today's prevention controls? 10 = almost certain. 5–6 = occasional. 1 = prevented by design, for example mistake-proofing.

D Detection, 1 to 10

How likely are today's checks to catch it before it leaves? 10 = no check at all. 5–6 = manual inspection. 1 = automatic check that can't miss.

Need a PFMEA your customer will sign off?

Grindstone builds and reviews PFMEAs and control plans with your team, for APQP and PPAP, at a fixed price agreed before we start.

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1 PFMEA worksheet

Sheet 1 of 5 · one row per failure mode, process steps in order

PROCESS STEP	FUNCTION / REQUIREMENT	FAILURE MODE	EFFECT	S	CAUSE	PREVENTION CONTROLS	O	DETECTION CONTROLS	D	RPN OR AP	RECOMMENDED ACTION	OWNER / DUE

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2 PFMEA worksheet

Sheet 2 of 5 · one row per failure mode, process steps in order

PROCESS STEP	FUNCTION / REQUIREMENT	FAILURE MODE	EFFECT	S	CAUSE	PREVENTION CONTROLS	O	DETECTION CONTROLS	D	RPN OR AP	RECOMMENDED ACTION	OWNER / DUE

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3 PFMEA worksheet

Sheet 3 of 5 · one row per failure mode, process steps in order

PROCESS STEP	FUNCTION / REQUIREMENT	FAILURE MODE	EFFECT	S	CAUSE	PREVENTION CONTROLS	O	DETECTION CONTROLS	D	RPN OR AP	RECOMMENDED ACTION	OWNER / DUE

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4 PFMEA worksheet

Sheet 4 of 5 · one row per failure mode, process steps in order

PROCESS STEP	FUNCTION / REQUIREMENT	FAILURE MODE	EFFECT	S	CAUSE	PREVENTION CONTROLS	O	DETECTION CONTROLS	D	RPN OR AP	RECOMMENDED ACTION	OWNER / DUE

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5 PFMEA worksheet

Sheet 5 of 5 · one row per failure mode, process steps in order

PROCESS STEP	FUNCTION / REQUIREMENT	FAILURE MODE	EFFECT	S	CAUSE	PREVENTION CONTROLS	O	DETECTION CONTROLS	D	RPN OR AP	RECOMMENDED ACTION	OWNER / DUE

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